

Problem 1.2.

Any number $\omega \in (0, 1]$ a non-terminating dyadic expansion $\omega = \sum_n d_n(\omega)2^{-n}$. Given $\varepsilon > 0$ take $N \in \mathbb{N}$ such that $2^{-N} < \varepsilon$. Now take

$$x_1 = \sum_{n=1}^N d_n(\omega)2^{-n} + \sum_{n=N+1}^{\infty} 2^{-n} \quad | \quad \text{dyadic expansion ending in all ones}$$

$$x_2 = \sum_{n=1}^N d_n(\omega)2^{-n} + \sum_{n=N+1}^{\infty} 2^{-2n} \quad | \quad \text{dyadic expansion ending in alternating ones and zeros}$$

The definitions immediately yield $x_1 \in N^c$ and $x_2 \in N$ and furthermore one has

$$|\omega - x_1| \leq \sum_{n=N+1}^{\infty} 2^{-n} = 2^{-N} < \varepsilon$$

$$|\omega - x_2| \leq \sum_{n=N+1}^{\infty} 2^{-n} = 2^{-N} < \varepsilon,$$

which yields the claim.